

Category Theory, Presheaf and Sheaf

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We cover the following topics in this note.

- Category
 - Functor
 - Natural Transformation
 - Determinant as Natural Transformation
 - Presheaf and Sheaf
 - TBA
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1 Category, Functor and Natural Transformation

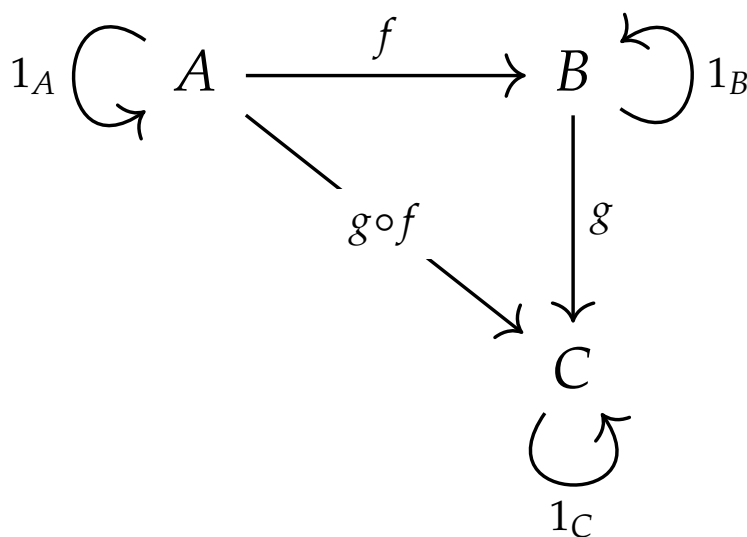


Figure 1: Category

Category

Definition 1. A **category** C consists of the following data:

- (i) **(Objects)** A class $\text{Ob}(C)$, whose elements are called **objects** of category C .
- (ii) **(Morphisms)** For each pair of objects $A, B \in \text{Ob}(C)$, a class $\text{Mor}_C(A, B)^a$, whose elements are called **morphisms** (or **arrows**) from A to B .
- (iii) **(Composition)** For any $f \in \text{Mor}_C(A, B)$ and $g \in \text{Mor}_C(B, C)$, a **composition**

$$(g \circ f) \in \text{Mor}_C(A, C).$$

These data are to satisfy the following axioms:

1. **(Identity)** For every object $A \in \text{Ob}(C)$, there exists an **identity morphisms**

$$1_A \in \text{Mor}_C(A, A).$$

2. **(Associativity)** For any $f \in \text{Mor}_C(A, B)$, $g \in \text{Mor}_C(B, C)$, and $h \in \text{Mor}_C(C, D)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

^aSometimes, we also denote it by $\text{Hom}(A, B)$ or $C(A, B)$.

Remark 1.

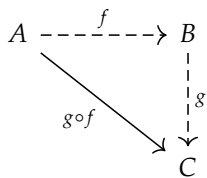


Figure 2: Composition of Morphisms

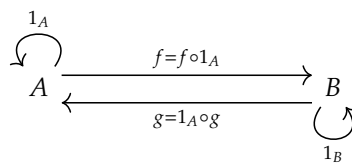


Figure 3: Identity Morphisms

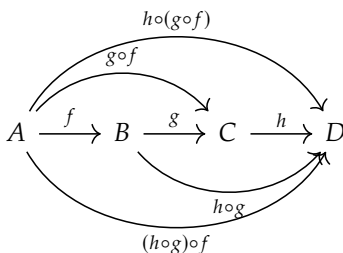


Figure 4: Associativity of Composition

Example 1.

Category	Object	Morphism
$\mathcal{S}et$	Sets	Functions between sets
$\mathcal{G}roup$	Groups	Group Homomorphisms
$\mathcal{R}ing$	Rings	Ring Homomorphisms
$\mathcal{V}ector\ Space$	Vector Spaces	Linear Transformations
$\mathcal{T}opology\ Space$	Topological Spaces	Continuous Maps
$\mathcal{P}oset$	Partially Ordered Sets	Order-preserving Maps
$\vdots$	$\vdots$	$\vdots$

Functor

Definition 2. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a mapping between two categories $\mathcal{C}$ and $\mathcal{D}$ is given by the following data:

- (i) **(Object Mapping)** A mapping that assign to each object $A \in \text{Ob}(\mathcal{C})$ an object $F(A) \in \text{Ob}(\mathcal{D})$:

$$\begin{aligned} F &: \text{Ob}(\mathcal{C}) \longrightarrow \text{Ob}(\mathcal{D}) \\ A &\longmapsto F(A) \end{aligned}$$

- (ii) **(Morphism Mapping)** For each pair $A, B \in \text{Ob}(\mathcal{C})$, a mapping that assigns to each morphism $\phi \in \text{Mor}_{\mathcal{C}}(A, B)$ a morphism $F(\phi) \in \text{Mor}_{\mathcal{D}}(F(A), F(B))$:

$$\begin{aligned} F &: \text{Mor}_{\mathcal{C}}(A, B) \longrightarrow \text{Mor}_{\mathcal{D}}(F(A), F(B)) \\ \phi &\longmapsto F(\phi) \end{aligned}$$

Remark 2 (Compatibility with Identity Morphisms and Composition). These functors satisfy the following conditions:

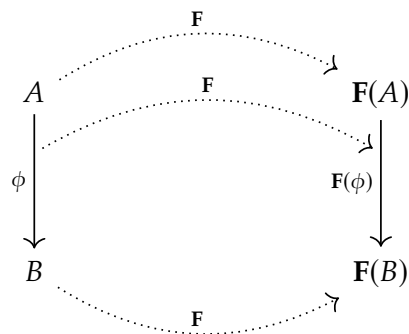
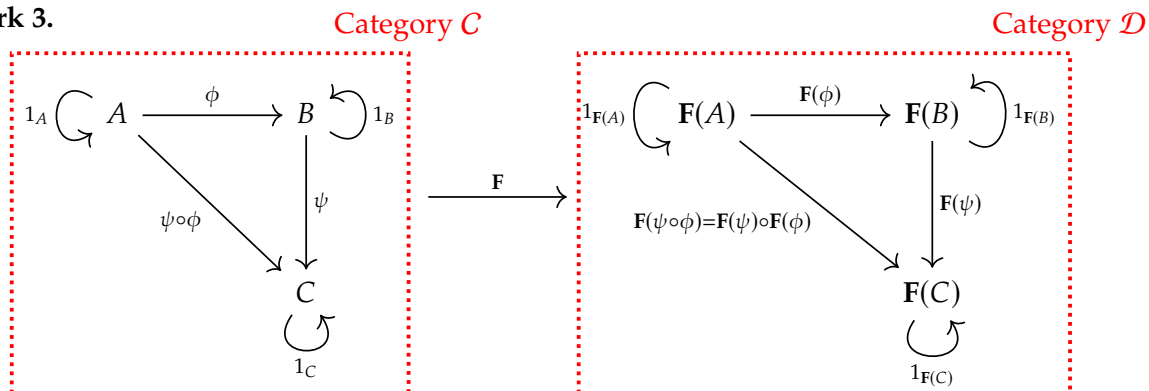
1. **(Preservation of Identity Morphisms)** For every object $A \in \text{Ob}(\mathcal{C})$,

$$F(1_A) = 1_{F(A)}.$$

2. **(Preservation of Composition)** For all morphisms $\phi \in \text{Mor}_{\mathcal{C}}(A, B)$ and $\psi \in \text{Mor}_{\mathcal{C}}(B, C)$,

$$F(\psi \circ \phi) = F(\psi) \circ F(\phi).$$

Remark 3.



Example 2.

Category (Source)	Functor	Object Mapping	Morphism Mapping	Category (Target)
TBA	TBA	TBA	TBA	TBA
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Natural Transformation

Definition 3. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and let $\mathbf{F}, \mathbf{G} : \mathcal{C} \rightarrow \mathcal{D}$ be two functors. A **natural transformation** (or a **morphism of functors**) $\eta : \mathbf{F} \Rightarrow \mathbf{G}$, is a collection $\{\eta_A\}_{A \in \text{Ob}(\mathcal{C})}$ such that

- (i) (**Component Morphisms**) For each object $A \in \text{Ob}(\mathcal{C})$, there is a morphism

$$\eta_A : \mathbf{F}(A) \rightarrow \mathbf{G}(A)$$

in the category $\mathcal{D}$, called the **component of η at A** .

- (ii) (**Naturality Condition**) For each morphism $\phi : A \rightarrow B$ in $\mathcal{C}$ the following diagram in $\mathcal{D}$ commutes:

$$\begin{array}{ccc} \mathbf{F}(A) & \xrightarrow{\eta_A} & \mathbf{G}(A) \\ \mathbf{F}(\phi) \downarrow & & \downarrow \mathbf{G}(\phi) \\ \mathbf{F}(B) & \xrightarrow{\eta_B} & \mathbf{G}(B) \end{array}$$

In other words,

$$\eta_B \circ \mathbf{F}(\phi) = \mathbf{G}(\phi) \circ \eta_A$$

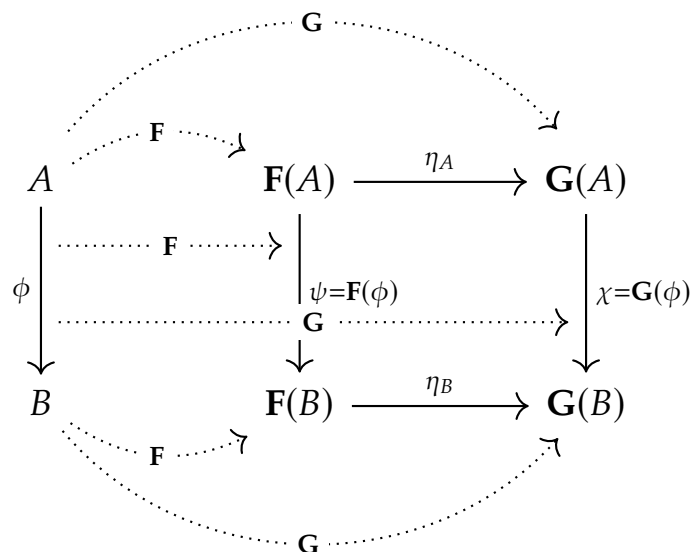
for all $\phi : A \rightarrow B$ in $\mathcal{C}$.

Remark 4. Sometimes we use the diagram

$$\begin{array}{ccc} & \mathbf{F} & \\ \curvearrowright & \Downarrow \eta & \curvearrowleft \\ \mathcal{C} & & \mathcal{D} \\ \curvearrowleft & \mathbf{G} & \end{array}$$

to indicate that η is a morphism from $\mathbf{F}$ to $\mathbf{G}$.

Remark 5.



Example 3 (Determinant). We start with the basic concepts and gradually introduce the determinant as a mathematical object.

Step 1. Understanding Matrices.

A matrix is a rectangular array of numbers. For example, a 2×2 matrix look like this:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where a, b, c, d are numbers.

Step 2. Invertibility of Matrices. The inverse of $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ (if it exists) is given by:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The **determinant** is a single number associated with a matrix, defined for 2×2 matrices as:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Step 3. The Determinant as a Function from $M_2(\mathbb{R})$ to $\mathbb{R}$

The set $M_2(\mathbb{R})$ is the set of all 2×2 matrices with real entries:

$$M_2(\mathbb{R}) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}.$$

The determinant is a function:

$$\begin{aligned} \det &: M_2(\mathbb{R}) \longrightarrow \mathbb{R} \\ A &\longmapsto \det(A) = ad - bc \in \mathbb{R} \end{aligned}$$

for each $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$. This function assigns a real number to each 2×2 matrix.

Step 4. Properties of the Determinant Function.

- **(Multiplicativity)** For any $A, B \in M_2(\mathbb{R})$,

$$\det(A * B) = \det(A) \cdot \det(B)$$

with $*$ is matrix multiplication and $\cdot$ is number multiplication.

- **(Normalization)** The determinant of the identity matrix is:

$$\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1.$$

Step 5. Group Structure

We identify a natural subset of $M_2(\mathbb{R})$ where:

- The determinant is nonzero.
- The operation of matrix multiplication is preserved.

1. (Closure Under Multiplication) For A, B in this subset, their product is also a 2×2 matrix. Using the determinant property,

$$\det(A) \neq 0 \wedge \det(B) \neq 0 \implies \det(AB) = \det(A) \det(B) \neq 0.$$

2. (Associativity) Matrix multiplication is associative.

3. (Existence of Identity and Inverses) The identity matrix $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ satisfies $\det(I) = 1 \neq 0$, and for every matrix A in this subset, there exists A^{-1} since $\det(A) \neq 0$.

These properties define a group, and this subset of matrices is precisely the general linear group of invertible matrices:

$$\mathrm{GL}(2, \mathbb{R}) := \{A \in M_2(\mathbb{R}), \det(A) = ad - bc \neq 0\}.$$

Step 6. The Determinant as a Group Homomorphism from $\mathrm{GL}(2, \mathbb{R})$ to $\mathbb{R}^\times$

The determinant is restricted to invertible matrices:

$$\det : \mathrm{GL}(2, \mathbb{R}) \rightarrow \mathbb{R}^\times,$$

where $\mathbb{R}^\times$ is the set of nonzero real numbers under multiplication. The determinant preserves the group operation because of the multiplicative property:

$$\det(A * B) = \det(A) \cdot \det(B).$$

for all $A, B \in \mathrm{GL}(2, \mathbb{R})$.

Step 7. Matrices Over a Commutative Ring

Let R be a commutative ring. The determinant of $A \in M_2(R)$ is define as

$$\det A = a \cdot d + -(b \cdot c),$$

using the addition and multiplication operations of the commutative ring R .

Step 8. Categories

- The Category CRing: The Category of Commutative Rings
 - Objects: commutative rings
 - Morphisms: ring homomorphisms
 - Composition: standard composition of functions
 - Identity Morphisms: $1_R : R \rightarrow R$ defined by $1_R(a) = a$ for $a \in R$
- The Category Grp: The Category of Groups
 - Objects: groups
 - Morphisms: group homomorphisms
 - Composition: standard composition of functions
 - Identity Morphisms: $1_G : G \rightarrow G$ defined by $1_G(a) = a$ for $a \in G$

Step 9. Functors

- The General Linear Group Functor $\mathbf{GL}_2$
 - $R \mapsto \mathbf{GL}_2(R)$
 - $(\phi : R \rightarrow S) \mapsto (\mathbf{GL}_2(\phi) : \mathbf{GL}_2(R) \rightarrow \mathbf{GL}_2(S))$
- The Unit Functor $\mathbf{U}$
 - $R \mapsto \mathbf{U}(R) = R^\times$
 - $(\phi : R \rightarrow S) \mapsto (\mathbf{U}(\phi) : R^\times \rightarrow S^\times)$

Step 10. Determinant as Natural Transformation

The determinant map is a family of group homomorphisms

$$\{\eta_R\}_{R \in \text{Ob}(\mathbf{CRing})}, \quad \text{where } \eta_R : \mathbf{GL}(2, R) \rightarrow R^\times,$$

one for each ring R , defined by

$$\eta_R(A) = \det A.$$

$$\begin{array}{ccccc} R & & GL_n(R) & \xrightarrow{\eta_R = \det_R} & R^\times \\ \downarrow \phi & & \downarrow GL_n(\phi) & & \downarrow U(\phi) \\ S & & GL_n(S) & \xrightarrow{\eta_S = \det_S} & S^\times \end{array}$$

$$\begin{array}{ccccc} & & \mathbf{U} & & \\ & \cdots & \downarrow & \cdots & \\ R & \xrightarrow{\quad GL_n \quad} & GL_n(R) & \xrightarrow{\det_R} & R^\times \\ \vdots & \cdots & \downarrow & \cdots & \downarrow \\ & \cdots & \mathbf{U} & \cdots & \\ S & \xrightarrow{\quad GL_n \quad} & GL_n(S) & \xrightarrow{\det_S} & S^\times \\ & \cdots & \downarrow & \cdots & \\ & & \mathbf{U} & & \end{array}$$

$\sigma = GL_2(\phi)$ (vertical arrow from $GL_2(R)$ to $GL_2(S)$)
 $\tau = U(\phi)$ (vertical arrow from $R^\times$ to $S^\times$)

2 Presheaf and Sheaf

Presheaf

Definition 4. Let X be a topological space. A **presheaf** $\mathcal{F}$ of abelian groups on X consists of

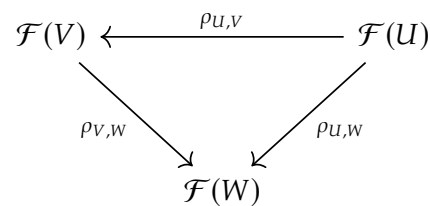
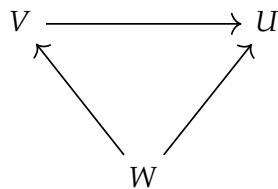
- (i) For each open $U \subseteq X$, an abelian group $\mathcal{F}(U)$, elements of $\mathcal{F}(U)$ are called sections of $\mathcal{F}$ over U .
- (ii) For each $V \subseteq U$ a homomorphism (called restriction map)

$$\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

such that $\rho_{U,U}$ is an identity and for open sets $W \subseteq V \subseteq U$ we have

$$\rho_{U,W} = \rho_{V,W} \circ \rho_{U,V}.$$

Remark 6.



Sheaf

Definition 5. TBA